

Describe the various renewable energy resources available in India and their potential

Renewable energy resources are natural energy sources that can be used again and again without getting exhausted. India has large potential for renewable energy because of its climate, rivers, sunlight, wind, and long coastal areas.

1. Solar Energy

Description

Solar energy is obtained from sunlight using solar panels and solar thermal systems.

Potential in India

- India receives sunlight for about 300 sunny days in a year.
- Rajasthan, Gujarat, Tamil Nadu, Andhra Pradesh, and Maharashtra have high solar potential.
- India has huge solar power generation capacity because of tropical climate.
- Solar parks are developed in many states.

Advantages

- Pollution free energy.
- Unlimited source of energy.
- Suitable for villages and cities.

Applications

- Solar street lights.
- Solar water heaters.
- Solar power plants.

2. Wind Energy

Description

Wind energy converts the energy of moving air into electrical energy using wind turbines.

Potential in India

- India is one of the leading countries in wind power generation.
- Tamil Nadu, Gujarat, Karnataka, Maharashtra, and Rajasthan have strong wind potential.

- Coastal regions and hilly areas are suitable for wind farms.

Advantages

- Renewable and clean energy.
- No fuel cost.
- Reduces air pollution.

Applications

- Electricity generation.
- Water pumping.
- Battery charging.

3. Biomass Energy

Description

Biomass energy is produced from agricultural waste, wood, animal waste, and organic materials.

Potential in India

- India is an agricultural country with large biomass resources.
- Crop residues, sugarcane waste, rice husk, and cow dung are widely available.
- Punjab, Haryana, Uttar Pradesh, and Maharashtra have high biomass potential.

Advantages

- Uses waste materials effectively.
- Reduces environmental pollution.
- Produces biogas and electricity.

Applications

- Biogas plants.
- Rural electricity generation.
- Industrial heating.

4. Geothermal Energy

Description

Geothermal energy uses heat from inside the earth to generate electricity.

Potential in India

- Geothermal resources are found in Himachal Pradesh, Ladakh, Jammu and Kashmir, and Gujarat.
- India has several hot springs suitable for geothermal projects.
- Puga Valley in Ladakh has high geothermal potential.

Advantages

- Continuous power generation.
- Less pollution.
- Requires less land.

Applications

- Electricity generation.
- Space heating.
- Industrial heating.

5. Hydroelectric Energy

Description

Hydroelectric energy generates electricity using flowing water from rivers and dams.

Potential in India

- India has many rivers and dams suitable for hydroelectric power.
- Himalayan rivers and Western Ghats provide large hydro potential.
- Himachal Pradesh, Uttarakhand, Kerala, and Karnataka have major hydroelectric projects.

Advantages

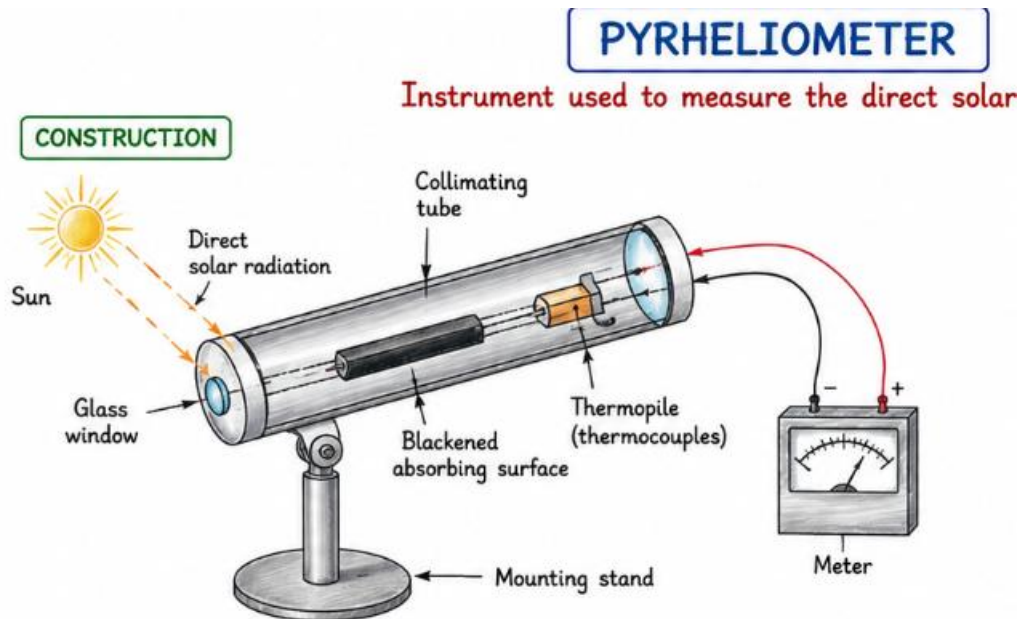
- Renewable and eco-friendly.
- Low operating cost.
- Reliable power generation.

Applications

- Electricity generation.
- Irrigation projects.
- Flood control.

Pyrheliometer

A **pyrheliometer** is an instrument used to measure **direct solar radiation** from the sun.



Construction

1. Long Tube

- A long tube allows only direct sun rays inside.

2. Glass Window

- Glass cover allows sunlight to enter.

3. Black Absorbing Plate

- Black surface absorbs maximum heat.

4. Thermopile

- Converts heat into electrical signal.

5. Meter

- Measures the electrical output.

Working Principle : It works on **thermoelectric effect**.

Working Steps

1. Instrument is pointed towards the sun.
2. Sun rays enter through the tube.

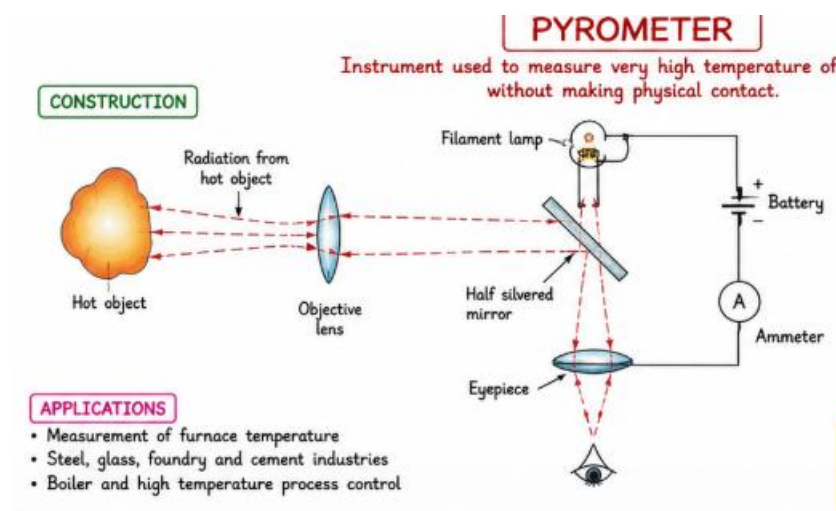
3. Black plate absorbs solar heat.
4. Thermopile gets heated.
5. Small electrical voltage is produced.
6. Meter shows solar radiation value.

Applications

- Solar power plants
- Weather stations
- Solar energy studies

Pyrometer

A **pyrometer** is an instrument used to measure **very high temperature without touching the object**.



Construction

1. Lens System

- Focuses heat radiation from hot object.

2. Filament Lamp

- Brightness is compared with hot body.

3. Eyepiece

- Used for viewing.

4. Battery

- Supplies current to filament.

5. Meter

- Shows temperature value.

Working Principle

Principle

Hot objects emit radiation according to temperature.

Working Steps

1. Pyrometer is aimed at hot object.
2. Radiation enters through lens.
3. Filament brightness is adjusted.
4. When filament disappears, brightness becomes equal.
5. Temperature is noted from meter scale.

Applications

- Furnace temperature measurement
- Steel industries
- Boilers

Difference Between Pyrheliometer and Pyrometer

Pyrheliometer	Pyrometer
Measures solar radiation	Measures high temperature
Used in solar studies	Used in industries
Pointed towards sun	Pointed towards hot object

Conclusion

- **Pyrheliometer** measures direct solar radiation.
- **Pyrometer** measures high temperature without contact.

